

C++ Increment and Decrement Operators

The increment operator ++ adds 1 to its operand, and the decrement operator -- subtracts 1 from its operand.

```
x = x+1;  
is the same as  
x++;  
can be written as:  
++x; // prefix form  
x++; // postfix form
```

```
x = x-1;  
is the same as  
x--;  
can be written as:  
--x; // prefix form  
x--; // postfix form
```

Example (1)

Following is the example to understand the difference between prefix and postfix form.

```
#include <iostream>  
using namespace std;  
main() {  
    int a = 21;  
    int c ;  
  
    // Value of a will not be increased before assignment.  
    c = a++;  
    cout << "Line 1 - Value of a++ is :" << c << endl ;  
  
    // After expression value of a is increased  
    cout << "Line 2 - Value of a is :" << a << endl ;  
  
    // Value of a will be increased before assignment.  
    c = ++a;  
    cout << "Line 3 - Value of ++a is :" << c << endl ;  
    return 0;  
}
```

Output:

```
Line 1 - Value of a++ is :21  
Line 2 - Value of a is :22  
Line 3 - Value of ++a is :23
```

If statement

The IF statement is used to express conditional expression. If the given condition is true then it will execute the statements.

Example (2)

Write a C++ program to read a number and check if it's positive, if it's so print it, add it to a total, and decrement it by 2:

```
#include <iostream>
using namespace std;

main() {
    int num, total=0;
    cout <<"please, input the number : ";
    cin >> num;
    if ( num >= 0)
    { cout << num <<" is a positive"<<endl;
      total += num;
      num = num - 2;
      cout<< "total= " <<total<<endl;
      cout<<"num= " <<num<<endl;
    }
    return 0;
}
```

Output:
please, input the number : 11
11 is a positive
total= 11
num= 9

Example (3):

Write a program to calculate a 10% discount if the amount purchased is greater than \$100, and no discount otherwise.

```
#include<iostream>
using namespace std;
int main() {
    int amount;
    float discount;
    cout<<"Input Amount of purchasing, please::";
    cin>> amount;
    cout<<"Amount of purchasing : " <<amount<<endl;

    if (amount >= 100)
    {
        discount = amount * 10 / 100;
        cout<<" The Discount is = " <<discount<<endl;
    }

    else
    {cout<<"Discount not applicable";}
    return 0;
}
```

Output:
Input Amount of purchasing,
please::350
Amount of purchasing : 350
The Discount is = 35

Laboratory tasks to groups A, B, C:

- 1- Develop the program in example (1) to perform prefix and postfix decrement?
- 2- Write program to compute the tax on employee's income. If the income is below 10000, the tax is applicable at 10%. For the income above 10000, the excess income is charged at 15%.